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10/15/24  
TMT Deliverable 2

### **S&P 500 Index**

The S&P 500 measures the total value of the market capitalizations of the largest 500 publicly traded companies divided by an index divisor. The S&P 500 is a free float-adjusted market-cap-weighted index, meaning it is constantly recalculated to adjust for the number of shares being traded. The weighted aspect means, the larger the company, the more weight it has in the index. Also, since the value of the 500 stocks included fluctuates over time, the S&P rebalances which stocks it includes each quarter. To qualify for the S&P 500, a company must meet the required market cap and have positive earnings for the previous four quarters. Data from July 31, 2020, shows the top four sectors by weight in the S&P 500 were Information Technology (27.48%), Health Care (14.58%), Consumer Discretionary (11.18%) and Communication Services (10.90%). On October 1, 2024, the top four sectors were Information Technology (32.40%), Financials (12.59%), Health Care (12.21%), and Consumer Discretionary (10.20%).

From March 2020 to the end of December 2021 the S&P 500 rose relatively steadily from \$2,840 to about \$4,744. It then experienced an overall decline to \$3,595 early in October 2022 before rising to \$5,840 at the present day (10/15/24). Although contractionary monetary policy caused interest rates to go up drastically from 2022-2024, the S&P 500 index didn't decrease or appear to slow with the change.

### **S&P 500 Volatility Index (VIX)**

The VIX measures the market's expectations for the near-term volatility of price changes in the S&P 500 index. More drastic price fluctuations in the S&P 500 show up as higher levels of volatility in the VIX. The VIX is useful to investors for decision-making as a measure of fear or risk in the market.

In the past 5 years, VIX had a peak of 65.9% in the middle of March 2020 from here it had a sharp drop until mid-August 2020 at 22.06%. It dipped slightly and fluctuated around 20% until March 2023, then fell more and has been fluctuating around about 15% until the present day (10/15/24). Since the VIX hasn't fluctuated too drastically high in the past few years, it has appeared to align with how the S&P 500 has been relatively steadily increasing. The exception was the VIX's spike in the beginning of March 2020 but I would expect this is related to the uncertainty in the market due to Covid-19.

### **US Treasury 5-Year Yield**

The 5-year treasury yield is the effective annual interest rate the US government pays an investor who purchases a 5-year treasury note. The price of treasury notes and the yield are inversely related – the yield falls when the price rises in secondary trading and vice versa.

The 5-year yield has risen significantly overall since March 2020. It had a slight decline from March 2020 to August 2020 which aligns with the interest rate drop in this period as treasury

yields typically go down when the Fed lowers interest rates. Then the 5-year yield rose gradually until August 2021 at 2.9%, and then more sharply until November 2022 at 3.9% where it's been fluctuating around until the present day. These increases also track with the increase in interest rates.

### **ICE BofA US High Yield OAS**

The ICE BofA Option-Adjusted Spreads (OAS) are the spreads between an OAS index of all bonds in the category of below-investment-grade rated corporate debt and a treasury curve. To be included in this index, a security must have a fixed coupon schedule, over one year of remaining maturity, and a \$100 million outstanding minimum. The ICE BofA OAS enables investors to more easily analyze whether an investment is worth its price by separating the cash flow from bonds and the cash flows of their embedded options (call options).

The ICE BofA OAS shot up from the beginning of March 2020 at 4.97% until its peak of 10.87% toward the end of March 2020 and has declined to around 3% in July 2021, rose to 5.93% by July 2022 but has declined until the present day (10/15/24) at 2.97%. Typically, the more volatile the interest rate, the lower the ICE BofA OAS. The rising ICE BofA OAS aligns with how the interest rate was stable for a while; then the declining ICE BofA OAS aligns with how the interest rate shifted to drastically rising.

### **US Dollar Index**

The US dollar index measures the value of the US dollar relative to six other currencies weighted in a basket: the euro (57.6%), Japanese yen (13.6%), British pound (11.9%), Canadian dollar (9.1%), Swedish krona (4.2%), and Swiss franc (3.6%).

The US dollar index was \$98.39 at the beginning of March 2020, rose to \$113.37 in October 2022, and fell slightly to where it has fluctuated around \$100 until 10/14/24. An index value of \$98.39 suggests that the value of the U.S. dollar depreciated 1.61% compared to the basket of currencies, and a value of \$113.37 suggests the U.S. dollar appreciated 13.37%. The U.S. dollar index is heavily influenced by inflation and/or deflation of the U.S. dollar and the currencies it is compared to.

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